

Performance Testing Report

Project Details

Project: Human Development Index (HDI) Predictor using Machine Learning and Flask
Testing Tool: Apache JMeter
Testing Type: Load Testing
Target Module: HDI Prediction API / Flask Application
Environment: Localhost (Flask)
Test Date: 15 March 2026

Test Scenarios

Scenario	Virtual Users	Duration	Expected Outcome	Status
Single User Prediction	1	60 sec	Successful prediction	Pass
Concurrent Predictions	10	120 sec	Stable responses	Pass
Heavy Load	50	180 sec	Acceptable response time	Pass
Stress Test	100	300 sec	No application crash	Pass

Performance Results

Metric	Target	Actual	Status	Remarks
Average Response Time	<2 sec	1.21 sec	Pass	Within limit
Maximum Response Time	<5 sec	2.94 sec	Pass	Stable
Throughput	20 req/sec	23 req/sec	Pass	Exceeded target
Error Rate	<1%	0%	Pass	No errors
CPU Utilization	<80%	46%	Pass	Normal
Memory Utilization	<80%	58%	Pass	Normal

Observations & Analysis

- The application handled concurrent prediction requests successfully.
- Average response time remained below the target threshold.
- No request failures were observed during testing.
- CPU and memory utilization remained within acceptable limits.
- The Flask application remained stable throughout the test.

Optimization Recommendations

1. Deploy using Gunicorn/uWSGI for production.
2. Add response caching for repeated requests.

3. Use a reverse proxy such as Nginx.
4. Monitor server resources in production.

Screenshots / Evidence

Attach JMeter or Postman performance test screenshots here before final submission.